

VENDOR PERFORMANCE & SUPPLY CHAIN ANALYTICS

Prepared By : Prashant Marathe

Tech Stack : Power BI | Power Query | DAX | Python | Pandas | NumPy | SQL | SQLite | Excel | GitHub

Core Skills : Data Cleaning | Data Modeling | ETL | KPI Development | Vendor Analytics | Procurement Analytics | Profitability Analysis | Business Intelligence

1. Executive Summary

This project develops an end-to-end Power BI analytics solution for evaluating vendor, product/brand, procurement, sales, and profitability performance. The workflow combines data-quality validation, Power Query preparation, dimensional modeling, DAX measure development, and interactive dashboard design.

The source dataset contains 10,648 records and 19 fields. Across the dataset, total sales are \$450,441,509.65, total purchases are \$320,911,160.31, and gross profit is \$129,530,349.34, producing an aggregate gross margin of 28.8%. The model covers 125 vendors and 10,648 distinct Brand codes.

2. Business Problem

Raw procurement and sales data does not directly answer which vendors drive revenue, where profit is generated, how purchasing compares with sales, or where inventory may require attention. The project converts transaction-level summary data into a reusable analytical model and a decision-support dashboard.

Key business questions:

- Which vendors contribute the most sales and gross profit?
- How do vendor sales compare with purchase spend?
- Which brands/product codes contribute the most sales and gross profit?
- How does gross margin vary across vendors and product codes?
- How does the sales-to-purchase dollar ratio vary across the portfolio?
- Which records require data-quality or inventory interpretation rather than simple imputation?

3. Dataset Overview

Metric	Value	Notes
Rows	10,648	Unique VendorNumber + Brand combinations
Columns	19	Vendor, product, pricing, quantity, sales, cost and KPI fields
Vendors	125	Distinct VendorNumber values
Brand codes	10,648	Brand is effectively unique at this dataset grain
Missing AverageSalesPrice	178	All associated with zero sales quantity
Missing GrossMarginPct	178	Legitimate where sales revenue is zero

4. Data Quality & Validation

Data profiling identified 178 rows with missing AverageSalesPrice and GrossMarginPct. These rows have zero sales quantity and zero sales dollars but contain purchase activity. Their gross profit equals the negative purchase dollars, indicating purchased but unsold inventory rather than a conventional missing-value problem. The project therefore preserves these nulls instead of replacing them with zero.

A VendorNumber + Brand duplicate check returned no duplicate combinations. Calculation checks were also used for purchase dollars, sales dollars, gross profit, and gross margin. Minor floating-point differences in derived currency calculations are treated as numerical precision artifacts.

There are 2,106 records with negative gross profit. This is a material analytical signal and should be interpreted at the product/vendor level rather than hidden through blanket cleaning.

5. Data Preparation & Modeling

Power Query

Imported the CSV, profiled the full dataset, reviewed completeness/distribution, standardized data types, and retained an auditable QA workflow.

Fact table

The primary analytical table was renamed FactVendorSales and contains the business measures and row-level attributes.

Vendor dimension

DimVendor was created as a reference table containing VendorNumber and VendorName, with duplicates removed.

Product dimension

DimProduct was created as a reference table for Brand and Description. Because Brand is unique at the current grain, the model treats the product/brand level carefully rather than forcing an ambiguous relationship.

Relationship

DimVendor[VendorNumber] → FactVendorSales[VendorNumber], 1:* cardinality, single-direction filtering, active relationship.

Filter validation

Vendor filtering was tested through vendor-level tables and measures to confirm that sales and gross profit change by vendor.

6. DAX Measures

Total Sales =
SUM(FactVendorSales[TotalSalesDollars])

Total Purchases =
SUM(FactVendorSales[TotalPurchaseDollars])

Gross Profit =
SUM(FactVendorSales[GrossProfit])

Gross Margin % =
DIVIDE([Gross Profit], [Total Sales], BLANK())

Total Purchase Qty =
SUM(FactVendorSales[TotalPurchaseQuantity])

Total Sales Qty =
SUM(FactVendorSales[TotalSalesQuantity])

Total Freight Cost =
SUM(FactVendorSales[FreightCost])

Total Excise Tax =
SUM(FactVendorSales[TotalExciseTax])

Vendor Count =
DISTINCTCOUNT(FactVendorSales[VendorNumber])

Brand Count =
DISTINCTCOUNT(FactVendorSales[Brand])

Sales to Purchase Ratio =
DIVIDE([Total Sales], [Total Purchases], BLANK())

7. Power BI Dashboard

Page 1 — Executive Overview

Executive KPI cards plus Top 10 vendor/brand sales and gross-profit views, and a vendor sales-versus-purchases comparison. Designed to answer the highest-level performance questions quickly.

Page 2 — Vendor Performance Analysis

Vendor Count, Sales, Purchases, Gross Profit and Gross Margin KPIs; Top 15 vendor comparisons; vendor gross margin and sales-to-purchase ratio views; and a detailed vendor table.

Page 3 — Product & Brand Performance

Brand Count, Sales, Purchases, Gross Profit and Gross Margin KPIs; Top 15 brand-code comparisons; margin and sales-to-purchase ratio views; and a detailed product/brand table.

8. Key Analytical Findings

- Aggregate sales are \$450.44M versus \$320.91M of purchases, resulting in \$129.53M gross profit and a 28.76% aggregate gross margin.
- The largest vendor by sales is DIAGEO NORTH AMERICA INC, with \$68.74M in sales, representing 15.3% of total sales.
- The top 10 vendors together account for 65.2% of total sales, indicating meaningful vendor concentration in the sales portfolio.
- Top vendor-level gross-profit contribution is led by DIAGEO NORTH AMERICA INC.
- There are 2,106 negative-gross-profit records, so profitability analysis should consider product/vendor-level exceptions rather than relying only on portfolio averages.
- The 178 zero-sales records represent \$282,183.65 of purchase spend; these records should be investigated as unsold inventory rather than imputed away.

9. Important Data Interpretation Notes

The dataset's Brand field has 10,648 distinct values across 10,648 rows. Consequently, Brand behaves more like a product/code identifier than a conventional repeated brand dimension in this extract. The report therefore avoids claiming broad brand-level aggregation beyond what the source structure supports.

The Sales-to-Purchase Ratio is defined here as total sales dollars divided by total purchase dollars. It is a dollar conversion/comparison ratio and should not be described as inventory turnover.

Gross Margin % is calculated as aggregate Gross Profit divided by aggregate Total Sales. This avoids averaging row-level margins, which can be distorted by different sales volumes and zero-sales records.

10. Tools & Technologies

Power BI Desktop • Power Query • DAX • Data Modeling • CSV • Data Profiling • KPI Design • Vendor Analytics • Procurement Analytics • Profitability Analysis

11. Challenges & Solutions

- Missing margin and average-sales-price values → investigated their business meaning and retained legitimate nulls.
- Potential duplicate-key risk → validated VendorNumber + Brand combinations before modeling.
- Filter-context validation → tested DimVendor filtering against sales and gross-profit measures.
- Aggregate margin calculation → used `DIVIDE([Gross Profit], [Total Sales])` rather than averaging row-level margins.
- Dashboard scope → separated executive, vendor, and product/brand analysis into focused report pages.

12. Conclusion

The completed Power BI solution demonstrates an end-to-end analytics workflow: source-data validation, Power Query preparation, dimensional modeling, DAX measure creation, interactive visual analysis, and business-oriented reporting. The project is suitable for a Data Analyst / BI Analyst portfolio because it shows both technical execution and the ability to translate operational data into structured business questions.

13. Future Enhancements

- Add time-series analysis if transaction dates become available.
- Add explicit inventory-aging or slow-moving-stock metrics using inventory snapshots or days-on-hand data.
- Introduce a validated product dimension if a stable product/brand master is available.
- Add drill-through pages for vendor and product investigations.
- Add row-level security and Power BI Service deployment for a production-style implementation.

14. Portfolio Positioning

Recommended project title: Vendor Performance & Supply Chain Analytics | Power BI. Recommended portfolio keywords: Power BI, DAX, Power Query, SQL, Python, Data Cleaning, Data Modeling, KPI Reporting, Vendor Analytics, Procurement Analytics, Profitability Analysis, Business Intelligence.